Name(s)_____ Period _____ Date _____

Activity Guide - Robot Face Planning

Sprite Scene Overview

Using shapes, variables, and randomNumbers, you're going to create a unique robot face. Be as creative as you can! Think about how you can make your robot unique in design; from the shape of its head, the number of eyes or mouths, or if it has antennas! There is no limit, so let's come up with something original and unexpected!

Before you jump into programming your robot face, you need to plan! Let's use our handy Problem-Solving Process to help us! You can refer back to the Problem Solving Process and Programming handout to help you with the steps!

Define

Write out a short description of your robot face, including what sorts of shapes and random features it will include.

_____

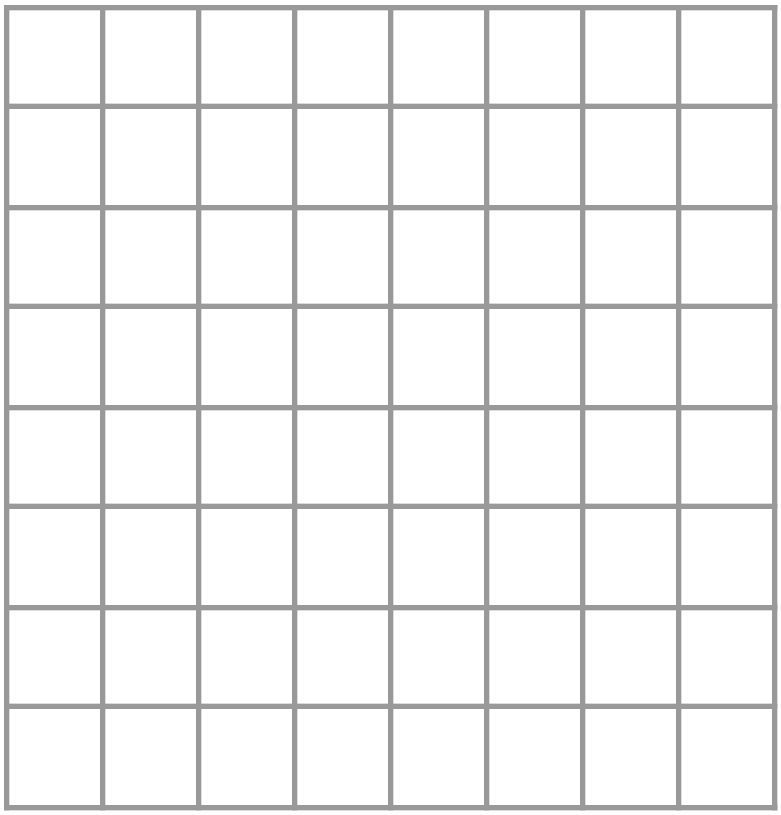
_____

_____

_____

Prepare

The space below is 50 by 50. That means each square on the paper will map to 50 pixels on the computer. Sketch out your **robot face using at least 4 different shape commands**. List the variables you'll use on the next page.



Shapes:
background(color)
rect(x, y, width, height)
ellipse(x, y, width, height)
line(x1, y1, x2, y2)
arc(x, y, width, height, start, stop)
point(x, y)
regularPolygon(x, y, sides, size)
shape(x1, y1, x2, y2, ... xN, yN)
fill("color")
noFill()
stroke("color")
noStroke()
strokeWeight()

List at least 4 variables you'll use to give your robot face random features below.

Variable Name	Description

Try

Once your teacher has approved your plan, go to Code Studio and start coding.

Reflect

Answer the following questions from the Problem Solving Process and Programming handout:

Compare your program to the defined problem … Were you successful in solving all aspects of the problem? How do you know?

_____

_____

What part of your animation are you most proud of?

_____

_____

What was the most challenging?

_____

_____

Are there any changes or improvements you can make to this program or another like it in the future?

_____

_____